

QUESTION ONE (3 Marks) START A NEW BOOKLET**MULTIPLE CHOICE:** Write the correct alternative on your writing paper.

1. Which of the following gives $|z| = 2\sqrt{5}$ and $\arg z = -\pi + \tan^{-1} 2$ **1**
- (A) $2 + 4i$
- (B) $-2 + 4i$
- (C) $-2 - 4i$
- (D) $2 - 4i$
-
2. Which of the following represent the three cubed roots of -8 **1**
- (A) $-2, 1 - \sqrt{3}i, 1 + \sqrt{3}i$
- (B) $-2, -1 - \sqrt{3}i, 1 + \sqrt{3}i$
- (C) $-2, 1 - \sqrt{3}i, 1 - \sqrt{3}i$
- (D) $2, 1 - \sqrt{3}i, 1 + \sqrt{3}i$
-
3. If z and w are the roots of the equation $3x^2 + (2 - i)x + (4 + i) = 0$, **1**
 which of the following represent $\frac{1}{z} + \frac{1}{w}$
- (A) $\frac{-7 - 6i}{17}$
- (B) $\frac{-7 + 6i}{15}$
- (C) $\frac{-7 - 6i}{15}$
- (D) $\frac{7 + 6i}{17}$

QUESTION TWO (23Marks)

- (a) Let $z = 3 + i$ and $w = 2 - 5i$. Find in the form $x + iy$,
- (i) z^2 1
- (ii) $\bar{z} + w$ 1
- (iii) $\frac{w}{z}$ 1
- (b) Find all pairs of integers x and y that satisfy $(x + iy)^2 = 24 + 10i$ 3
- (c) Sketch the region in the complex plane where the inequalities 3
- $$|z - 1 - i| \leq 2 \text{ and } 0 < \arg(z - 1 - i) < \frac{\pi}{4}$$
- hold simultaneously.
- (d) (i) Write $1 + i$ in the form $r(\cos \theta + i \sin \theta)$ 2
- (ii) Hence, or otherwise, find $(1 + i)^{17}$ in the form $x + iy$, 3
where a and b are integers.
- (e) Prove by Mathematical Induction that $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ 4
for all integers $n \geq 1$.
- (e) If ω is the non real cube root of 1,
- (i) Prove that $(1 + 2\omega + 3\omega^2)(1 + 3\omega + 2\omega^2) = 3$. 2
- (ii) Prove that $(1 + 2\omega + 3\omega^2) + (1 + 3\omega + 2\omega^2) = -3$ 1
- (iii) Hence find the exact values of $(1 + 2\omega + 3\omega^2)$ and $(1 + 3\omega + 2\omega^2)$. 2

QUESTION THREE (28 Marks) START A NEW BOOKLET

- (a) Draw separate sketches of the following, clearly indicating any critical points, asymptotes, discontinuities etc. Make each sketch about half a page.

(i) $y = (x+1)(x-1)$ 1

(ii) $y = \frac{1}{(x+1)(x-1)}$ 2

(iii) $y = \sqrt{(x+1)(x-1)}$ 2

(iv) $y^2 = (x+1)(x-1)$ 2

(v) $y = |(x+1)(x-1)|$ 2

(vi) $y = [(x+1)(x-1)]^2$ 2

(vii) $|y| = (x+1)(x-1)$ 2

(viii) $y = \log_e (x+1)(x-1)$ 2

(b) For the function $f(x) = \frac{x^4}{x^2-1}$.

(i) Determine whether $f(x)$ is odd, even or neither. 1

(ii) Find the coordinates of the stationary points. 3

(iii) By considering large values of $|x|$ and any discontinuities, sketch the graph of $f(x) = \frac{x^4}{x^2-1}$ showing all essential features. 4

(c) The equation of a curve is given by $3x^2 + y^2 - 2xy - 8x + 2 = 0$.

(i) Find $\frac{dy}{dx}$. 2

(ii) Find the coordinates of the points on the curve where the tangent to the curve is parallel to the line $y = 2x$. 3

QUESTION FOUR (3 Marks) START A NEW BOOKLET**MULTIPLE CHOICE: Write the correct alternative on your writing paper**

1. When $P(x) = x^4 - 1$ is factorised over the complex field it may be written as
 - (A) $P(x) = (x^2 - 1)(x^2 + 1)$
 - (B) $P(x) = (x - 1)(x + 1)(x^2 + 1)$
 - (C) $P(x) = (x - 1)(x + 1)(x + i)(x - i)$
 - (D) $P(x) = (x^2 - 1)(x - i)^2$

2. If α, β, γ are the roots of the equation $x^3 - 8x^2 - 4x + 12 = 0$, then the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ is
 - (A) $\frac{8}{3}$
 - (B) $1 + \frac{1}{-8} + \frac{1}{-12}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{1}{-12}$

3. Given $\frac{1}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$, the values of A and B respectively are
 - (A) 1, -1
 - (B) -1, 1
 - (C) $\frac{1}{4}, -\frac{1}{4}$
 - (D) $-\frac{1}{4}, \frac{1}{4}$

QUESTION FIVE (23Marks) START A NEW BOOKLET

- (a) The equation $x^3 - 4x^2 + 5x + 2 = 0$ has roots α, β and γ .
- Find (i) $\alpha^2 + \beta^2 + \gamma^2$ 2
- (ii) $\alpha^3 + \beta^3 + \gamma^3$ 3
- (b) Factorise $3x^4 + 17x^3 + 30x^2 + 12x - 8 = 0$ completely if it has a root of multiplicity 3. 3
- (c) (i) Show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$. 2
- (ii) Find the roots of the equation $8x^3 - 6x - 1 = 0$ in terms of $\cos \theta$. 2
- (iii) Hence evaluate $\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$. 2
- (d) If $\frac{2x+31}{(x+1)^3(x+2)} = \frac{A}{(x+1)} + \frac{B}{(x+1)^2} + \frac{C}{(x+1)^3} + \frac{D}{(x+2)}$ 4
- find the values of A, B, C and D.
- (e) The roots of $x^3 + 3px + q = 0$ are α, β and γ (none of which are equal to 0).
- (i) Find the monic equation with roots $\frac{\beta\gamma}{\alpha}, \frac{\alpha\gamma}{\beta}$ and $\frac{\alpha\beta}{\gamma}$, giving the coefficients in terms of p and q . 3
- (ii) Deduce that if $\gamma = \alpha\beta$ then $(3p - q)^2 + q = 0$. 2

END OF PAPER

Question 1

$$1. |z| = 2\sqrt{5} \quad \arg z = -\pi + \tan^{-1} 2$$

z lies in the 3rd quadrant

$$\therefore z = -2 - 4i \quad (C)$$

$$2. \text{ Let } z^3 = -8$$

$$z^3 + 8 = 0$$

$$(z+2)(z^2-2z+4) = 0$$

$$(z+2)[(z-1)^2+3] = 0$$

$$(z+2)(z-1+\sqrt{3}i)(z-1-\sqrt{3}i) = 0$$

$$\therefore z = -2, 1-\sqrt{3}i, 1+\sqrt{3}i \quad (A)$$

$$3. 3x^2 + (2-i)x + (4+i) = 0$$

$$\alpha + \beta = -\frac{b}{a}$$

$$= -\frac{2+i}{3}$$

$$\alpha\beta = \frac{c}{a}$$

$$= \frac{4+i}{3}$$

$$\frac{1}{z} + \frac{1}{w} = \frac{w+z}{wz}$$

$$= \frac{-\frac{2+i}{3}}{\frac{4+i}{3}}$$

$$= \frac{-2+i}{4+i}$$

$$= \frac{-2+i}{4+i} \times \frac{4-i}{4-i}$$

$$= \frac{-8+6i-i^2}{17}$$

$$= \frac{-7+6i}{17}$$

$$\therefore \frac{1}{z} + \frac{1}{w} = \frac{-7-6i}{17} \quad (A)$$

Question 2

$$a) z = 3+i \quad w = 2-5i$$

$$\begin{aligned} \text{ii) } z^2 &= (3+i)^2 \\ &= 9+6i+i^2 \\ &= 8+6i \end{aligned}$$

$$\begin{aligned} \text{iii) } \bar{z} + w &= 3-i+2-5i \\ &= 5-6i \end{aligned}$$

$$\begin{aligned} \text{iii) } \frac{w}{z} &= \frac{2-5i}{3+i} \\ &= \frac{2-5i}{3+i} \times \frac{3-i}{3-i} \\ &= \frac{6-17i+5i^2}{10} \\ &= \frac{1}{10} - \frac{17}{10}i \end{aligned}$$

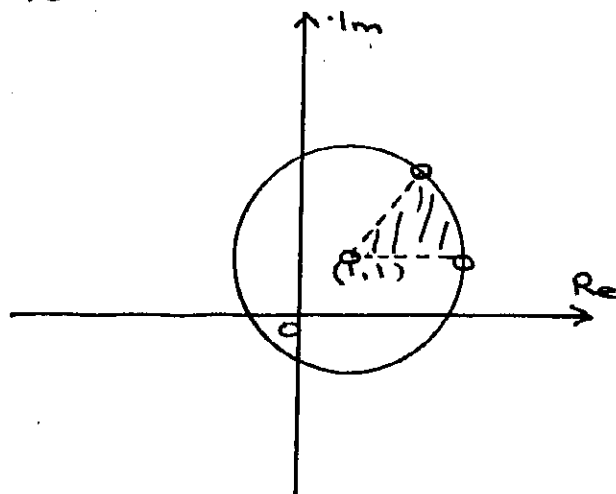
$$b) (x+iy)^2 = 24+10i$$

$$x^2 - y^2 = 24$$

$$xy = 5$$

$\therefore$ The 2 square roots are $\pm(5+i)$

$$c) |z-1-i| \leq 2 \quad 0 < \arg(z-1-i) < \frac{\pi}{4}$$



d) i) $z = 1 + i$

$$|z| = \sqrt{1^2 + 1^2} \\ = \sqrt{2}$$

$$\arg z = \tan^{-1} 1 \quad -\pi < \theta \leq \pi \\ = \frac{\pi}{4}$$

$$\therefore 1 + i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$(1 + i)^{17} = (\sqrt{2})^{17} \left(\cos \frac{17\pi}{4} + i \sin \frac{17\pi}{4} \right)$$

$$= 256\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$= 256 + 256i$$

e) Show $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta) \quad n \geq 1$

Step 1 Test $n=1$

$$\text{L.H.S} = (\cos \theta + i \sin \theta)^1$$

$$= \cos \theta + i \sin \theta$$

$$\text{R.H.S} = \cos \theta + i \sin \theta$$

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore$ Result is true for $n=1$

Step 2. Assume that the result is true for $n=k$
that is $(\cos \theta + i \sin \theta)^k = \cos(k\theta) + i \sin(k\theta)$

Step 3 Hence show the result is true for $n=k+1$
that is $(\cos \theta + i \sin \theta)^{k+1} = \cos(k+1)\theta + i \sin(k+1)\theta$

$$\text{L.H.S} = (\cos \theta + i \sin \theta)^{k+1}$$

$$= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta)$$

$$= (\cos k\theta + i \sin k\theta) (\cos \theta + i \sin \theta) \text{ by step 2}$$

$$= \cos k\theta \cos \theta + i \sin \theta \cos k\theta + i \sin k\theta \cos \theta \\ + i^2 \sin \theta \sin k\theta$$

$$= (\cos k\theta \cos \theta - \sin k\theta \sin \theta) +$$

$$i(\cos k\theta \sin \theta + \sin k\theta \cos \theta)$$

$$= \cos(k+1)\theta + i \sin(k+1)\theta$$

$$= \text{R.H.S}$$

Step 4 Since the result is true for $n=1$ then

from Step 3 it is true for $n=1+1=2$,
and then for $n=3$, and so on by the
process of mathematical induction it is true for
all positive integral values of n

$$e) \quad \omega^3 = 1$$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega - 1 \neq 0 \quad \therefore \omega^2 + \omega + 1 = 0$$

$$(i) \quad (1 + 2\omega + 3\omega^2)(1 + 3\omega + 2\omega^2)$$

$$= 1 + 3\omega + 2\omega^2 + 2\omega + 6\omega^2 + 4\omega^3 + 3\omega^2 + 9\omega^3 + 6\omega^4$$

$$= 1 + 5\omega + 11\omega^2 + 13\omega^3 + 6\omega^4$$

$$= 1 + 5\omega + 11\omega^2 + 13 + 6\omega$$

$$= 14 + 11\omega + 11\omega^2$$

$$= 14 + 11(\omega^2 + \omega)$$

$$= 14 - 11$$

$$= 3$$

$$\text{NB } ① \omega^3 = 1$$

$$② \omega^4 = \omega$$

$$③ \omega^2 + \omega + 1 = 0$$

$$\omega^2 + \omega = -1$$

$$(ii) \quad 1 + 2\omega + 3\omega^2 + 1 + 3\omega + 2\omega^2 = 2 + 5\omega + 5\omega^2$$

$$= 2 + 5(\omega + \omega^2)$$

$$= 2 - 5$$

$$= -3$$

(iii) From parts (i) and (ii) $(1 + 2\omega + 3\omega^2)$ and $(1 + 3\omega + 2\omega^2)$ are roots of $x^2 + 3x + 3 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3 \pm \sqrt{3^2 - 4 \times 1 \times 3}}{2 \times 1}$$

$$= \frac{-3 \pm \sqrt{3}}{2}$$

$$\therefore 1 + 2\omega + 3\omega^2 = \frac{-3 - \sqrt{3}}{2}$$

$$1 + 3\omega + 2\omega^2 = \frac{-3 + \sqrt{3}}{2}$$

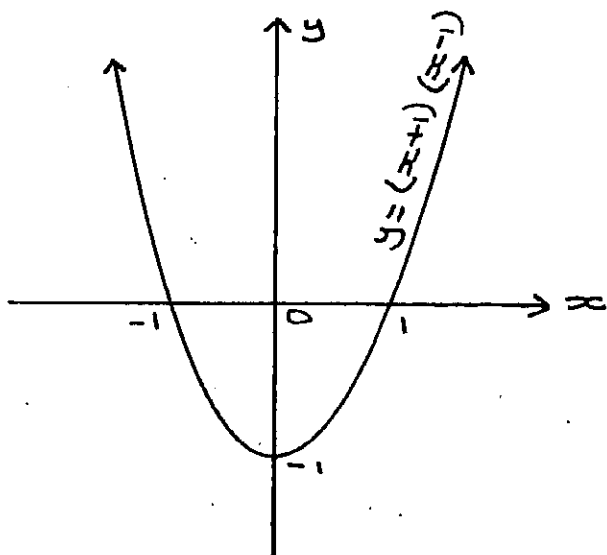
$$\text{NB } \operatorname{Im}(\omega) = -\operatorname{Im}(\omega^2)$$

$$\operatorname{Im}(2\omega + 3\omega^2) = \operatorname{Im}(\omega^2) < 0$$

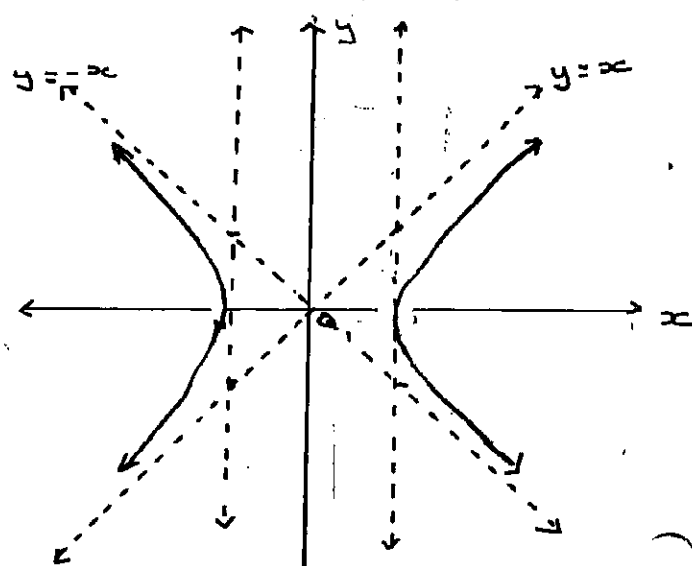
$$\operatorname{Im}(3\omega + 2\omega^2) = \operatorname{Im} \omega > 0$$

Question 3

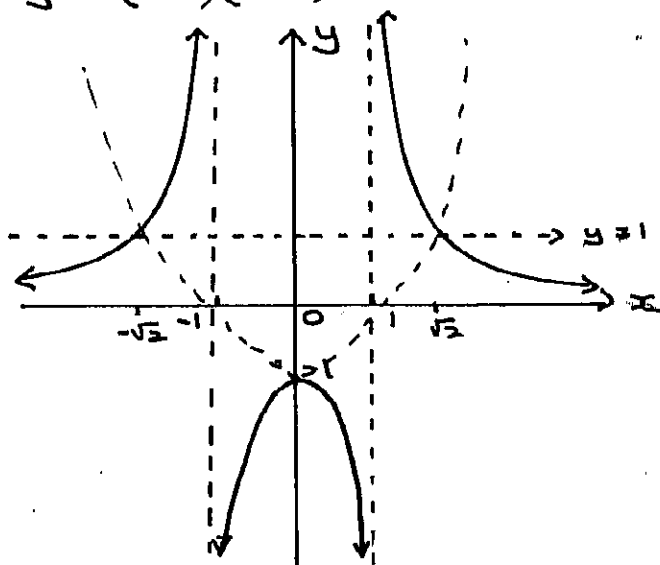
a) (i) $y = (x+1)(x-1)$



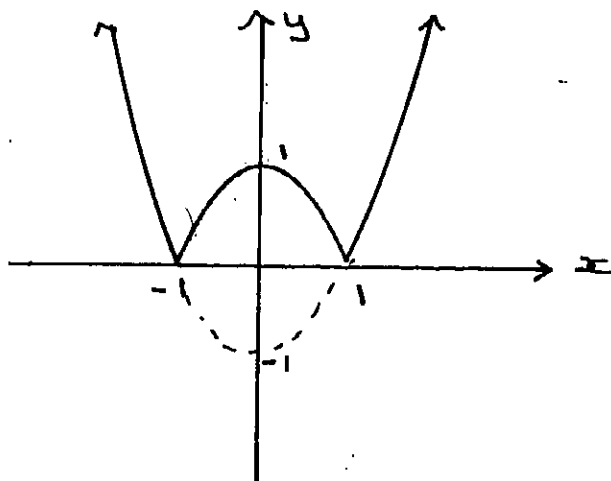
(v) $y^2 = (x+1)(x-1)$



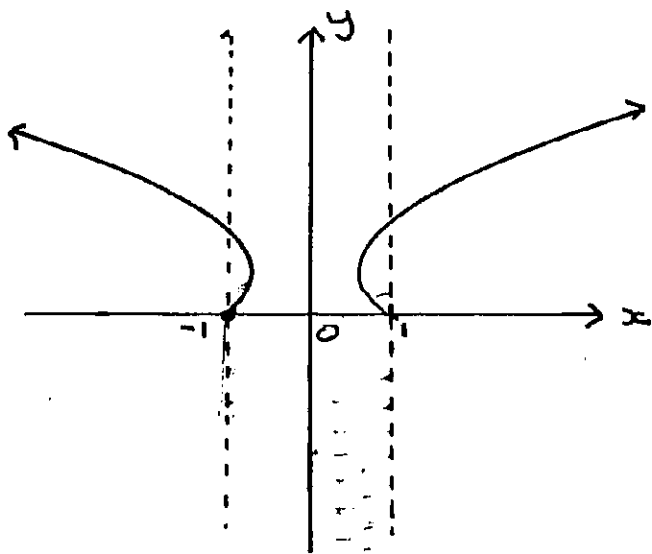
ii) $y = \frac{1}{(x-1)(x+1)}$



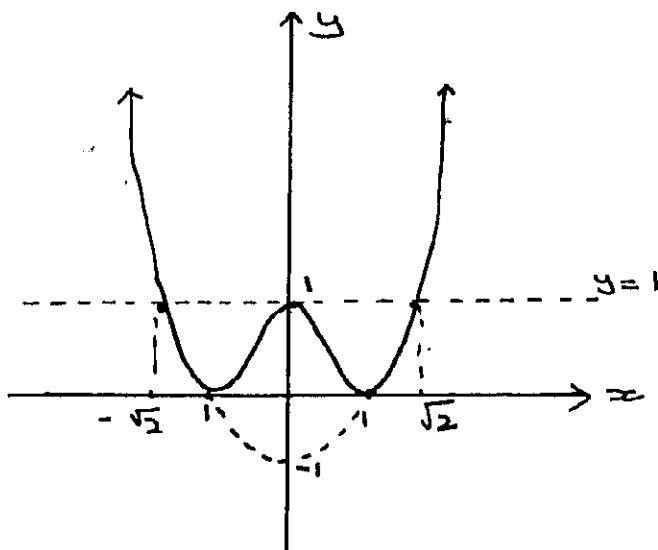
v) $y = |(x-1)(x+1)|$



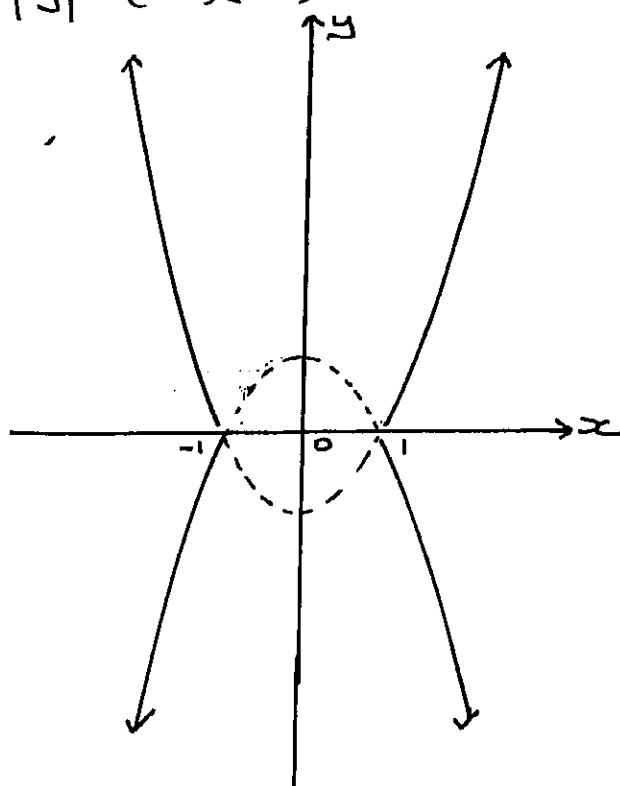
iii) $y = \sqrt{(x-1)(x+1)}$



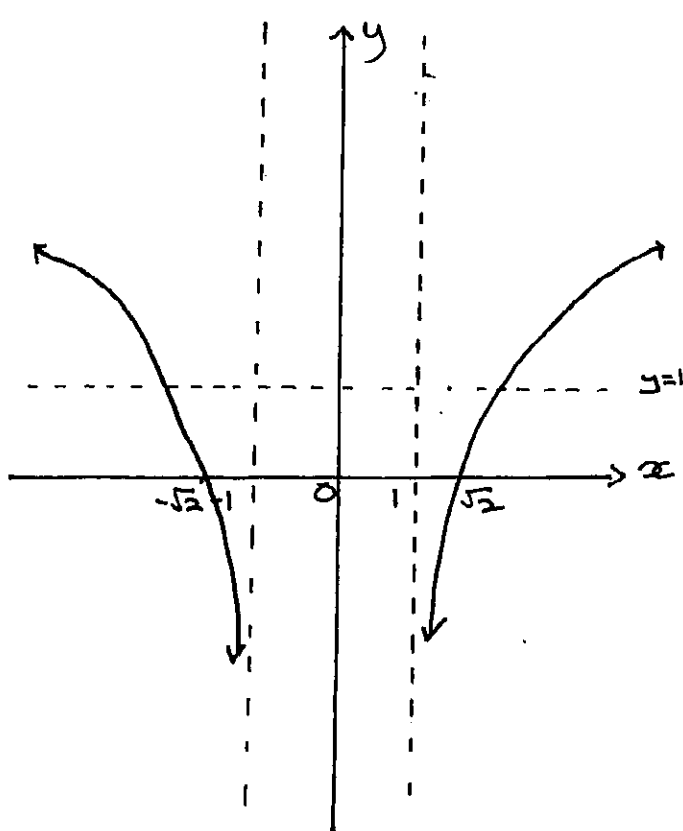
vi) $y = [(x+1)(x-1)]^2$



vii) $|y| = (x-1)(x+1)$



viii) $y = \log_e (x+1)(x-1)$



b) $f(x) = \frac{x^4}{x^2-1}$

$$f(-x) = \frac{(-x)^4}{(-x)^2-1} = \frac{x^4}{x^2-1}$$

$$\therefore f(x) = f(-x)$$

$\therefore f(x) = \frac{x^4}{x^2-1}$ is an even function.

ii) $f(x) = \frac{x^4}{x^2-1}$

$$u = x^4 \quad v = x^2-1$$

$$\frac{du}{dx} = 4x^3 \quad \frac{dv}{dx} = 2x$$

$$f'(x) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(x^2-1)4x^3 - x^4 \times 2x}{(x^2-1)^2}$$

$$= \frac{2x^3(2x^2-2-x^2)}{(x^2-1)^2}$$

$$= \frac{2x^3(x^2-2)}{(x^2-1)^2}$$

$$= \frac{2x^3(x-\sqrt{2})(x+\sqrt{2})}{(x^2-1)^2}$$

For stationary points

$$f'(x) = 0$$

$$\therefore 2x^3(x-\sqrt{2})(x+\sqrt{2}) = 0$$

$$\therefore x = 0, \pm\sqrt{2}$$

$$f(x) = \frac{x^4}{x^2-1}$$

$$f(0) = 0$$

$$f(\sqrt{2}) = \frac{(\sqrt{2})^4}{(\sqrt{2})^2-1}$$

$$= 4$$

$$f(-\sqrt{2}) = \frac{(-\sqrt{2})^4}{(-\sqrt{2})^2-1}$$

$$= 4$$

∴ The stationary points are (0,0) (√2, 4) and (-√2, 4)

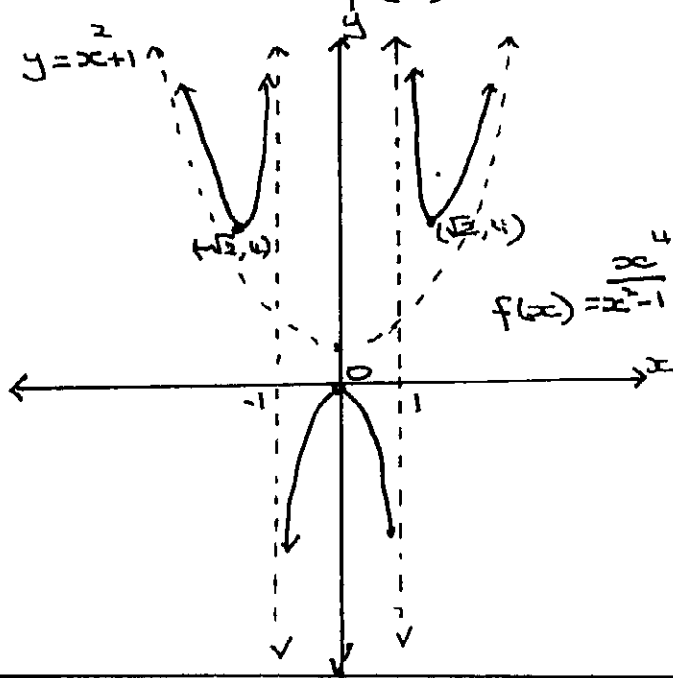
(iii) vertical asymptotes are $x = \pm 1$

$$f(x) = \frac{x^4}{x^2-1}$$

$$= x^2 + 1 + \frac{1}{x^2-1}$$

$$\text{as } x \rightarrow \infty \quad f(x) \rightarrow x^2 + 1$$

$$\text{as } x \rightarrow -\infty \quad f(x) \rightarrow x^2 + 1$$



$$c) \quad 3x^2 + y^2 - 2xy - 8x + 2 = 0 \quad \text{--- ①}$$

$$6x + 2y \frac{dy}{dx} - (2x \frac{dy}{dx} + 2y) - 8 = 0$$

$$(2y - 2x) \frac{dy}{dx} = 8 + 2y - 6x$$

$$\frac{dy}{dx} = \frac{8 + 2y - 6x}{2y - 2x}$$

$$\therefore \frac{dy}{dx} = \frac{4 + y - 3x}{y - x}$$

(ii) As the tangent is parallel to $y = 2x$, $\frac{dy}{dx} = 2$

$$\frac{4 + y - 3x}{y - x} = 2$$

$$4 + y - 3x = 2y - 2x$$

$$y = 4 - x \quad \text{--- ②}$$

sub ② in ①

$$3x^2 + y^2 - 2xy - 8x + 2 = 0$$

$$3x^2 + (4-x)^2 - 2x(4-x) - 8x + 2 = 0$$

$$3x^2 + 16 - 8x + x^2 - 8x + 2x^2 - 8x + 2 = 0$$

$$6x^2 - 24x + 18 = 0$$

$$x^2 - 4x + 3 = 0$$

$$(x-3)(x-1) = 0$$

$$x = 3, 1$$

when $x = 3$

$$\text{sub ② } y = 4 - x$$

$$= 4 - 3$$

$$= 1$$

when $x = 1$

$$\text{sub ② } y = 4 - x$$

$$= 4 - 1$$

$$= 3$$

∴ The points are (3, 1) (1, 3)

Question 4

$$\begin{aligned} 1. \quad P(x) &= x^4 - 1 \\ &= (x^2 - 1)(x^2 + 1) \\ &= (x-1)(x+1)(x-i)(x+i) \end{aligned} \quad (C)$$

$$2. \quad x^3 - 8x^2 - 4x + 12 = 0$$

$$\begin{aligned} \alpha + \beta + \gamma &= \frac{-b}{a} \\ &= 8 \end{aligned}$$

$$\begin{aligned} \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ &= -4 \end{aligned}$$

$$\begin{aligned} \alpha\beta\gamma &= \frac{-d}{a} \\ &= -12 \end{aligned}$$

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \alpha\gamma + \beta\gamma}{\alpha\beta\gamma} \\ &= \frac{-4}{-12} \\ &= \frac{1}{3} \quad (C) \end{aligned}$$

$$3. \quad \frac{1}{(x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{x-3}$$

$$1 = A(x-3) + B(x+1)$$

$$\text{Sub } x=3$$

$$1 = B(3+1)$$

$$B = \frac{1}{4}$$

$$\text{Sub } x=-1$$

$$1 = A(-1-3)$$

$$A = -\frac{1}{4}$$

$$\therefore A = -\frac{1}{4}, B = \frac{1}{4} \quad (D)$$

Question 5

$$a) \quad x^3 - 4x^2 + 5x + 2 = 0$$

$$\begin{aligned} \alpha + \beta + \gamma &= \frac{-b}{a} \\ &= 4 \end{aligned}$$

$$\begin{aligned} \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ &= 5 \end{aligned}$$

$$\begin{aligned} \alpha\beta\gamma &= \frac{-d}{a} \\ &= -2 \end{aligned}$$

$$\begin{aligned} i) \quad \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 4^2 - 2 \times 5 \\ &= 6 \end{aligned}$$

$$(ii) \quad \alpha^3 = 4\alpha^2 - 5\alpha - 2 \quad \text{--- (1)}$$

$$\beta^3 = 4\beta^2 - 5\beta - 2 \quad \text{--- (2)}$$

$$\gamma^3 = 4\gamma^2 - 5\gamma - 2 \quad \text{--- (3)}$$

$$\textcircled{1} + \textcircled{2} + \textcircled{3}$$

$$\begin{aligned} \alpha^3 + \beta^3 + \gamma^3 &= 4(\alpha^2 + \beta^2 + \gamma^2) - 5(\alpha + \beta + \gamma) - 2 \times 3 \\ &= 4(6) - 5(4) - 6 \\ &= -2 \end{aligned}$$

$$b) \quad 3x^4 + 17x^3 + 30x^2 + 12x - 8 = 0$$

$$P(x) = 3x^4 + 17x^3 + 30x^2 + 12x - 8$$

$$P'(x) = 12x^3 + 51x^2 + 60x + 12$$

$$P''(x) = 36x^2 + 102x + 60$$

$$= 2(18x^2 + 51x + 30)$$

$$= 2(x+2)(18x+15)$$

$$P(-2) = 0$$

$$P'(-2) = 0$$

$$P''(-2) = 0$$

$\therefore (x+2)$ is a triple root

$$\therefore 3x^4 + 17x^3 + 30x^2 + 12x - 8 = 0$$

$$(x+2)^3(3x-1) = 0$$

by inspection

c) By De Moivre's Theorem $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

$$i) (\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

$$\begin{aligned} \text{L.H.S} &= (\cos \theta + i \sin \theta)^3 \\ &= \cos^3 \theta + 3\cos^2 \theta i \sin \theta + 3\cos \theta (i \sin \theta)^2 + i^3 \sin^3 \theta \\ &= \cos^3 \theta + 3\cos^2 \theta i \sin \theta - 3\cos \theta \sin^2 \theta - i \sin^3 \theta \\ &= \cos^3 \theta - 3\cos \theta \sin^2 \theta + i(3\cos^2 \theta \sin \theta - \sin^3 \theta) \end{aligned}$$

Equating real parts

$$\begin{aligned} \cos 3\theta &= \cos^3 \theta - 3\cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta) \\ &= \cos^3 \theta - 3\cos \theta + 3\cos^3 \theta \end{aligned}$$

$$\therefore \cos 3\theta = 4\cos^3 \theta - 3\cos \theta$$

$$ii) 8x^3 - 6x - 1 = 0$$

$$4x^3 - 3x = \frac{1}{2}$$

$$\text{But } \cos 3\theta = \frac{1}{2}$$

$$\text{NB } 3\theta = \pm \frac{\pi}{3} + 2n\pi$$

n is integral

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$\therefore$ The roots of $8x^3 - 6x - 1 = 0$ are $\cos \frac{\pi}{9}$,

$$\cos \frac{5\pi}{9} = -\cos \frac{4\pi}{9} \text{ and } \cos \frac{7\pi}{9} = -\cos \frac{2\pi}{9}$$

$$iii) 8x^3 - 6x - 1 = 0$$

$$\begin{aligned} \text{product of the roots} &= \frac{-d}{a} \\ &= \frac{1}{8} \end{aligned}$$

$$\therefore \cos \frac{\pi}{9} \times -\cos \frac{4\pi}{9} \times -\cos \frac{2\pi}{9} = \frac{1}{8}$$

$$\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$$

$$d) \frac{2x+31}{(x-1)^3(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x-1)^3} + \frac{D}{(x+2)}$$

$$2x+31 = A(x-1)^2(x+2) + B(x-1)(x+2) + C(x+2) + D(x-1)^3$$

$$\underline{\text{Let } x=1} \quad 2(1)+31 = C(1+2)$$

$$33 = 3C$$

$$C = 11$$

$$\underline{\text{Let } x=-2} \quad 2(-2)+31 = D(-2-1)^3$$

$$27 = -27D$$

$$D = -1$$

$$\underline{\text{Let } x=0} \quad 2(0)+31 = A(-1)^2(2) + B(-1)(2) + 11(2) - 1(-1)^3$$

$$31 = 2A - 2B + 22 + 1$$

$$8 = 2A - 2B$$

$$4 = A - B \quad \text{--- (1)}$$

$$\underline{\text{Let } x=-1} \quad 2(-1)+31 = A(-1-1)^2 + B(-1-1)(-1+2) + 11(-1+2) - 1(-1-1)^3$$

$$29 = 4A - 2B + 11 + 8$$

$$10 = 4A - 2B$$

$$5 = 2A - B \quad \text{--- (2)}$$

Solve (1) and (2) simultaneously

$$A - B = 4 \quad \text{--- (1)}$$

$$2A - B = 5 \quad \text{--- (2)}$$

$$\textcircled{1} - \textcircled{2}$$

$$-A = -1$$

$$A = 1$$

Sub (1)

$$A - B = 4$$

$$1 - B = 4$$

$$B = -3$$

$$\therefore A = 1, B = -3, C = 11, D = -1$$

$$e) \quad x^3 + 3px + q = 0$$

$$\alpha + \beta + \gamma = -\frac{b}{a} \\ = 0$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} \\ = 3p$$

$$\alpha\beta\gamma = -\frac{d}{a} \\ = -q$$

$$\begin{aligned} \text{sum of the roots} &= \frac{\alpha\beta}{\gamma} + \frac{\beta\gamma}{\alpha} + \frac{\alpha\gamma}{\beta} \\ &= \frac{\alpha^2\beta^2 + \beta^2\gamma^2 + \alpha^2\gamma^2}{\alpha\beta\gamma} \\ &= \frac{(\alpha\beta + \beta\gamma + \alpha\gamma)^2 - 2\alpha\beta\gamma(\alpha + \beta + \gamma)}{\alpha\beta\gamma} \\ &= \frac{(3p)^2 - q(0)}{-q} \\ &= \frac{9p^2}{-q} \end{aligned}$$

$$\begin{aligned} \text{sum of the roots 2 at a time} &= \frac{\alpha\beta}{\gamma} \times \frac{\beta\gamma}{\alpha} + \frac{\beta\gamma}{\alpha} \times \frac{\gamma\alpha}{\beta} + \frac{\gamma\alpha}{\beta} \times \frac{\alpha\beta}{\gamma} \\ &= \beta^2 + \gamma^2 + \alpha^2 \\ &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 0 - 2(3p) \\ &= -6p \end{aligned}$$

$$\begin{aligned} \text{Product of the roots} &= \frac{\alpha\beta}{\gamma} \times \frac{\alpha\gamma}{\beta} \times \frac{\beta\gamma}{\alpha} \\ &= -q \end{aligned}$$

$$\therefore \text{The monic equation is } x^3 + \frac{9p^2}{q}x - 6px + q = 0$$

(ii) If $\gamma = \alpha\beta$ then one of the roots, $\frac{\alpha\beta}{\gamma} = \frac{\gamma}{\gamma} = 1$

$$\therefore x^3 + \frac{qp^2}{q}x^2 - 6px + q = 0$$

$$1 + \frac{qp^2}{q} - 6p + q = 0$$

$$q + qp^2 - 6pq + q^2 = 0$$

$$q + (3p - q)(3p - q) = 0$$

$$\therefore (3p - q)^2 + q = 0$$